

Absolutely! Below is a **detailed, step-by-step guide** to set up your **Dell PowerEdge R640** with:

-  **2 × SSDs** in **RAID 1** → for **Proxmox or ESXi OS**
-  **2 × HDDs** in **RAID 1** → for **VM storage / data**
-  Clean installation of **Proxmox VE** (recommended) or **ESXi**

This guide assumes:

- You have 4 drives installed (2 SSDs, 2 HDDs)
- You're using the built-in **Dell PERC H730P/H740P** RAID controller (common in R640)
- You want a **reliable, production-ready virtualization server**

 Step-by-Step Setup Guide: Dell R640 with RAID 1 + Proxmox

 Step 1: Prepare Your Tools

What You Need:

1. **USB flash drive (8GB+)** — to create bootable installer
2. **Proxmox VE ISO**:
<https://www.proxmox.com/en/downloads>
 - Or **ESXi ISO** (if preferred): Use Dell's custom version from [Dell Support](<https://www.dell.com/support>)

3. **Keyboard & Monitor** (or use iDRAC remotely)
4. **iDRAC access** (highly recommended — allows remote ISO mounting)

🛠️ Step 2: Enter RAID Configuration (PERC Utility)

> ⚠️ This will **erase all data** on the drives.

1. **Power on** the Dell R640.
2. When you see the Dell logo, press **Ctrl+R** repeatedly.
 - ➤ This enters the **PERC Configuration Utility** (RAID BIOS).
 - Wait a few seconds — it may not appear immediately.

🛠️ Step 3: Create RAID 1 for SSDs

In PERC Utility:

1. Use arrow keys to select **Controller 1** (PERC H730P/H740P).
2. Press **F2** → choose **Create Virtual Disk**.

Configure SSD RAID 1:

- **RAID Level**: Select **RAID 1** (Mirror)
- **Select Physical Disks**:
 - Use the **↑↓** arrows to highlight your **two SSDs**

- Press **Spacebar** to select both
- **Strip Size**: ` 64 KB` (default, fine)
- **Virtual Disk Name**: ` SSD-OS-MIRROR` (optional but helpful)
- **Read Policy**: ` Read Ahead`
- **Write Policy**: ` Write Back with BBU` (faster; safe if cache protected)
- **IO Policy**: ` Direct IO`
- **Access Policy**: ` Read/Write`
- **Disk Cache Policy**: ` Enabled`

 Press **OK** to confirm.

>  The system will ask: **"This operation will erase all data. Proceed?"** → **Yes**

Wait a few seconds — the virtual disk is created.

 Step 4: Create RAID 1 for HDDs

Repeat the process:

1. Press **F2** → **Create Virtual Disk**
2. **RAID Level**: ` RAID 1`
3. **Select Physical Disks**:
 - Choose your **two HDDs** (use Spacebar)
4. **Virtual Disk Name**: ` HDD-DATA-MIRROR`

5. Same settings as above:

- Strip: `64 KB`
- Write Policy: `Write Back with BBU`
- Rest: defaults

✓ Press **OK** → Confirm data loss.

Now you have:

- `SSD-OS-MIRROR` → will host Proxmox/ESXi
- `HDD-DATA-MIRROR` → will store VMs, backups, ISOs

🛠 Step 5: Save & Exit RAID Utility

1. Press **Esc** → choose **Home** or **Adapter Properties**
2. Press **F10** → **Apply Changes and Exit**
3. Server will reboot.

🛠 Step 6: Boot from Proxmox Installer (USB)

Create Bootable USB (Beforehand)

On another PC:

1. Download **Proxmox VE ISO**

2. Use **Rufus** (Windows) or `dd` (Linux/macOS) to write ISO to USB

> Example `dd` command (Linux/macOS):

```
` `` `bash
```

```
sudo dd if=proxmox-ve_*.iso of=/dev/sdX bs=1M status=progress && sync
```

```
` `` `
```

(Replace `/dev/sdX` with your USB drive)

Boot R640 from USB

1. Insert USB.
2. Power on.
3. Press **F11** during boot → **One-Time Boot Menu**
4. Select **USB Drive** (e.g., "Sandisk Cruzer" or "UEFI: USB")

> 💡 Or use **iDRAC Virtual Console** → "Virtual Media" → mount ISO remotely.

🛠 Step 7: Install Proxmox VE

Welcome Screen

- Select **Install Proxmox VE**
- Press **Enter**

Installation Wizard

1. **Choose Country, Language, Keyboard**

- Example:

- Country: United States
- Time Zone: New York
- Keyboard: US

→ Click **Next**

2. **Data Storage

- **Target Hard Disk**: Should show your **SSD RAID array** (e.g., `/dev/sda``)
- ☒ Check: *Use all space* (or customize later)

→ **Next**

3. **Type of Installation

- ☒ **Standard system installation**

→ **Next**

4. **Network Configuration

Fill in:

- **Hostname**: `pve01`` (or any name)
- **IP Address**: e.g., `192.168.1.100``
- **Netmask**: `255.255.255.0``

- **Gateway**: `192.168.1.1`
- **DNS**: `1.1.1.1` or your router IP

>  Make sure IP doesn't conflict with other devices.

→ **Next**

5. **Admin Password & Email**

- **Password**: Set a strong one
- **Email**: `admin@yourdomain.com` (for alerts)

→ **Next**

6. **Confirm & Install**

- Review settings
- Click **Install**

Wait 5–10 minutes. The system installs to the **SSD RAID 1 array**.

Step 8: Reboot & Remove USB

When installation completes:

- Screen says: **"Installation finished. Reboot now."**
- Press **Enter**

- Remove USB when prompted

Server reboots into Proxmox.

🛠️ Step 9: Access Proxmox Web Interface

Once booted:

- Look at screen for IP address (e.g., `https://192.168.1.100:8006`)
- Open a browser on another PC
- Go to: `https://<your-server-ip>:8006`

> 🔑 Log in with:

- **User**: `root@pam`
- **Password**: the one you set

✅ You're in!

🛠️ Step 10: Add HDD RAID as Storage (Optional but Recommended)

Now use the **HDD RAID 1 array** for VMs and backups.

In Proxmox Web UI:

1. Click your node (e.g., `pve01`) → **Storage** → **Add** → **Directory**

2. Fill in:

- **ID**: `hdd-data`
- **Directory**: `/mnt/pve/hdd-data`
- **Content**:
 - ☒ Disk image
 - ☒ Container
 - ☒ ISO images
 - ☒ Backup

3. Click **Create**

Mount the HDD RAID:

Back in shell (SSH or console):

```
` `` bash
```

```
# Format HDD RAID (only once — erases data!)
```

```
sudo mkfs.ext4 /dev/sdb1
```

```
# Create mount point
```

```
sudo mkdir /mnt/pve/hdd-data
```

```
# Mount
```

```
sudo mount /dev/sdb1 /mnt/pve/hdd-data
```

```
# Add to fstab for auto-mount
```

```
echo '/dev/sdb1 /mnt/pve/hdd-data ext4 defaults 0 0' | sudo tee -a /etc/fstab
```

` ` `

> 💡 Replace `/dev/sdb1` with correct device (check with `lsblk`)

Refresh Proxmox UI → storage should now show available space.

✅ Done! Your Server Is Ready

Now You Can:

- Create **VMs** (Ubuntu, Windows, etc.)
- Run **LXC containers** (lightweight Linux apps)
- Store **ISOs, backups, disks** on HDD RAID
- Enable **snapshots, backups, clustering** later

🔄 Optional: Install ESXi Instead?

If you prefer **ESXi**:

1. Download **Dell Custom ESXi 7.0/8.0 ISO** from [Dell Support](https://www.dell.com/support)
2. Follow **Steps 1–4** (RAID setup) above
3. Boot from ESXi USB

4. Install to ****SSD RAID 1 array****

5. Use ****HDD RAID 1**** as a datastore (via Storage in vSphere Client)

> Note: ESXi free version lacks backup/migration tools — consider Veeam or similar.

📌 Final Tips

- 🗝️ ****Enable SSH**** (Proxmox) or ****ESXi Shell**** for remote management
- 💾 ****Back up regularly**** — RAID is not a backup!
- 🔄 ****Update firmware**** via Dell Lifecycle Controller (F10 at boot)
- 💻 Use ****iDRAC**** for remote access (set static IP in iDRAC settings)

Need Help With?

Let me know:

- Whether you want ****ZFS instead of RAID****
- If you want ****dual-boot Proxmox + ESXi****
- How to set up ****Windows VMs, pfSense, NAS, etc.****

I'll give you exact commands and configurations! 🛠️